

Calculus I Lesson 21C

1) $y = f(x) = \sin^2(5x)$

Hint: $\sin^2(5x) = [\sin(5x)]^2$

a) $D_x(\sin 5x) = \underline{\hspace{2cm}}?$

b) $y' = f'(x) = D_x([\sin(5x)]^2) = 2 \cdot [\sin(5x)]^1 \cdot D_x(\sin 5x) = \underline{\hspace{2cm}}?$

2) $y = f(x) = \tan^3(x + 4)$

Hint: $\tan^3(x + 4) = [\tan(x + 4)]^3$

a) $D_x(\tan(x + 4)) = \underline{\hspace{2cm}}?$

b) $y' = f'(x) = D_x([\tan(x + 4)]^3) = 3 \cdot [\tan(x + 4)]^2 \cdot D_x(\tan(x + 4)) = \underline{\hspace{2cm}}?$

3) $y = f(x) = \cos^4(5x)$

Hint: $\cos^4(5x) = [\cos(5x)]^4$

a) $D_x(\cos(5x)) = \underline{\hspace{2cm}}?$

b) $y' = f'(x) = D_x([\cos(5x)]^4) = 4 \cdot [\cos(5x)]^3 \cdot D_x(\cos(5x)) = \underline{\hspace{2cm}}?$

4) $y = f(x) = (5x^3 + 7x)^4$

a) $D_x(5x^3 + 7x) = \underline{\hspace{2cm}}?$

b) $y' = f'(x) = D_x\left((5x^3 + 7x)^4\right) = 4 \cdot (5x^3 + 7x)^3 \cdot D_x(5x^3 + 7x) = \underline{\hspace{2cm}}?$

$$5) \ y = f(x) = (x^{1/3} + x)^5$$

$$a) \ D_x(x^{1/3} + x) = \underline{\hspace{2cm}}?$$

$$b) \ y' = f'(x) = D_x\left((x^{1/3} + x)^5\right) = 5 \cdot (x^{1/3} + x)^4 \cdot D_x(x^{1/3} + x) = \underline{\hspace{2cm}}?$$

6) $y = f(x) = (2x^3 + 4x)^6$

a) $D_x(2x^3 + 4x) = \underline{\hspace{2cm}}?$

b) $y' = f'(x) = D_x\left((2x^3 + 4x)^6\right) = 6(2x^3 + 4x)^5 \cdot D_x(2x^3 + 4x) = \underline{\hspace{2cm}}?$

7) $y = f(x) = \sqrt[4]{x^2 + 8}$

Hint: $\sqrt[4]{x^2 + 8} = (x^2 + 8)^{1/4}$

a) $D_x(x^2 + 8) = \underline{\hspace{2cm}}?$

b) $y' = f'(x) = D_x\left((x^2 + 8)^{1/4}\right) = \frac{1}{4} \cdot (x^2 + 8)^{1/4-1} \cdot D_x(x^2 + 8) = \underline{\hspace{2cm}}?$

8) $y = f(x) = \sqrt[3]{4x^2 + 3}$

Hint: $\sqrt[3]{4x^2 + 3} = (4x^2 + 3)^{1/3}$

a) $D_x(4x^2 + 3) = \underline{\hspace{2cm}}?$

b) $y' = f'(x) = D_x\left((4x^2 + 3)^{1/3}\right) = \frac{1}{3}(4x^2 + 3)^{1/3-1} \cdot D_x(4x^2 + 3) = \underline{\hspace{2cm}}?$

